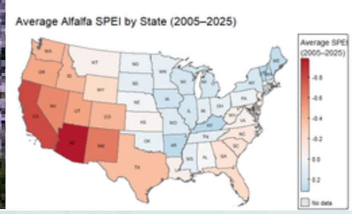
# Conserving Water in U.S. Alfalfa Production

## Implications of Drought & Smart Irrigation in Alfalfa System




AAEA - GSS & C-FARE 2026 Policy Communications
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## Scope of the Problem and Context:

- Alfalfa: 3rd. valuable filed crop, and occupied around 23.5 million acres in the U.S., which is key feed for dairy and beef.
- Irrigation: Concentrated and intensive in western states. Irrigated alfalfa consumes ≈15% of California's agricultural water and 26% of Colorado River water.
- Export: 20% alflafa exported to Asian countries from western states with raising demand of dairy.
- Environment/water policy scientists: Alfalfa delivers relatively low water-use efficiency returns (dollars/jobs per drop). When this hay is exported, the region is essentially exporting its scarce water
- Agronomists: alfalfa can survive "deficit irrigation", farmers can readily idle (fallow) stands or reduce irrigation under drought. And alfalfa provides agriculture ecosystem services, stable sales to farms, and less labor input required.

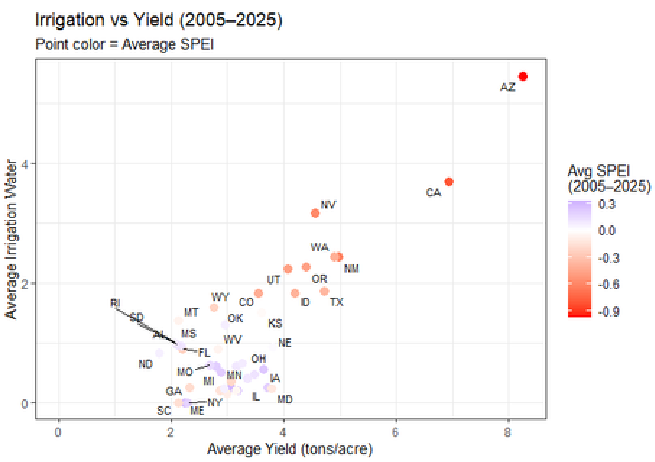
## Know the Facts:

### *Drought impacts on alfalfa*

Standard Precipitation-Evaporation Index (SPEI) is a drought index that measures the difference between water supply and demand.

Alfalfa is central to water debates in the West due to higher drought pressure and intensive irrigation demand.

Droughts significantly reduce alfalfa yields; irrigation mitigates losses.



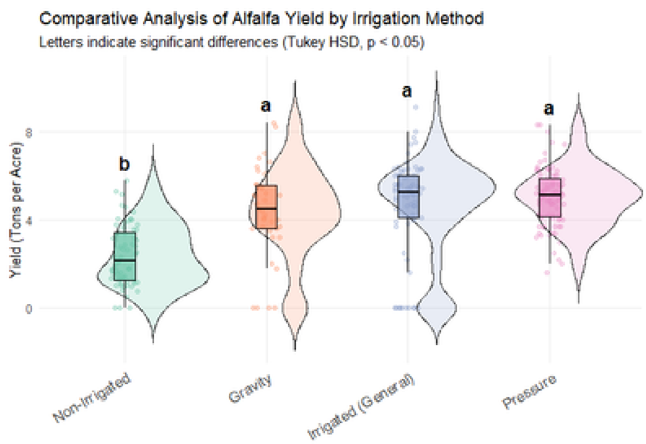

### *Alfalfa Market*

Sales contract and water rights allocation practices left many growers with little incentive to change growing practices.

Alfalfa export value show weaker association with drought pressure when consider irrigation and market price.

### *Irrigation methods result in different yields*

Pressure irrigation exhibits the lowest variability and the highest ton-per-acre-foot efficiency, underscoring irrigation's role as a key drought adaptation—while also highlighting exposure to tightening water constraints.



# Policy Options & Recommendations

### *1. IRRIGATION REGULATIONS*

Water conservation policies should protect both water resources and farm viability.

Reform water-rights systems by introducing volumetric water allocations, tiered pricing, and drought periods flexibility. This encourages producers to conserve water while still protecting farm income, forage production, and regional food-security needs.

### *2. IRRIGATION PRACTICE*

Provide cost-share incentives, technical training, and extension support for pressure irrigation, smart drip systems, soil-moisture sensors, and AI-based irrigation scheduling. These tools can reduce water waste, lower labor needs, and maintain alfalfa yield under semi-arid conditions.

For alfalfa growth, the soil moisture should be controled between 30% and 40% to achaive optimal irrigation efficiency. The automatic pressure irrigation will be trigered on when temperature higher than 30 celsus and humdity lower than 30%.

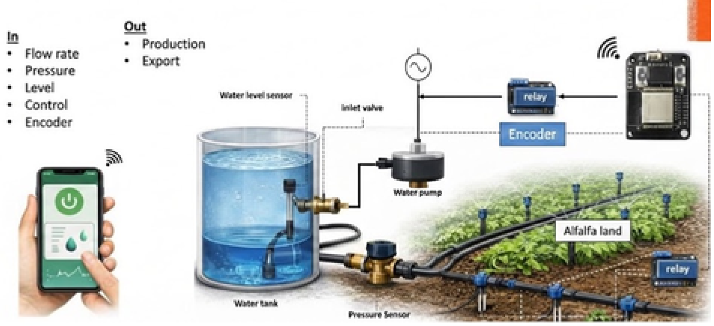
**Key thresholds**

- Soil moisture LOW 30%
- Soil moisture HIGH 50%
- Temperature >= 30 °C
- Humidity <= 30%
- bias +3

### *3. WATER ALLOCTION BUDGET*

Set basin-wide withdrawal caps based on groundwater availability, surface-water supply, drought risk, and crop demand. Planning should include farmers, water districts, scientists, environmental groups, agri-businesses, and export-market stakeholders to balance conservation, production, and trade.

**Objective:** Keep input water level maintain production and export




## Why AI-based smart irrigation system?

1. Flexible and precise irrigation volume according to real-time soil moisture, crop biocharacters, water availability, and microclimate conditions.
2. Less labor input and in time irrigation once evaporation higher than precipitation.
3. Water saving and higher efficiency because accurate irrigation to water absorb area of the crop.
4. Use the historical irrigation data, which uploaded in cloud platform (ThingSpeak) and analyzed by supervised machine learning in Matlab, to train the irrigation system and set better irrigation decision.

More appropriate for semi-arid agriculture, where production depends heavily on efficient water use, avoiding both water stress and water waste due to non-precise irrigation practices (e.g. gravity irrigation, fixed irrigation schedule, groundwater depletion).

**1 Sense** — Read soil moisture, OpenWeather prediction, temperature, and humidity on a repeating interval.

**2 Decide** — Compute LOW / HIGH thresholds and shift them earlier when it is hot and dry.

**3 Actuate** — Pump turns ON below LOW and OFF above HIGH - a classic hysteresis loop.

**4 Protect** — Low water level, sensor faults, max runtime, and minimum off-time all force safe behavior.

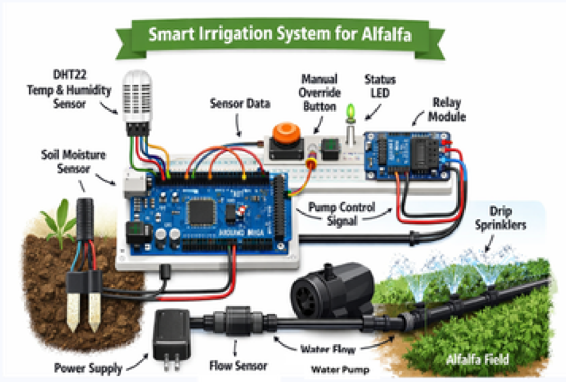
Read inputs and make discision → Auto/Manual Irrigation → Calibration and upload data to cloud

## Takeaways

Policy should connect farm-level technology with basin-level water planning. Smart irrigation data can support more transparent water allocation, drought response, and groundwater usage.

Cloud data from ThingSpeak can support better irrigation decisions and long-term water conservation.

Alfalfa's deficit-irrigation tolerance makes it suitable for flexible water-saving strategies during drought.

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